

YEAR 7 • UNIT 1 • HOMEWORK 2

Signs and Animal Cells

Mathematics 1.2 – Multiplying & Dividing Integers

Science 1.2 – Animal Cells

Name: _____

Class: _____

Date given: _____

Date due: _____

How to do this homework

- Write your answers **on this sheet**, in the spaces given. If you need more room, use the back of the page.
- In maths, **write the calculation before the answer**. Work out the **sign** first – it is half of every answer on this sheet.
- In science, answer in **full English sentences**. “Nucleus” is not a sentence; “The nucleus is the control centre of the cell” is.
- If a word is new to you, look for the blue **Word help** box on that page before you give up.

Section A • Mathematics 1.2 – Multiplying and Dividing Integers

Remember from the lessonDividing **undoes** multiplying, so $\times$ **and** $\div$ **obey the same four rules:****Same** signs the answer is **positive** $-6 \times -7 = 42$ $-42 \div -7 = 6$ **Different** signs the answer is **negative** $-6 \times 7 = -42$ $-42 \div 7 = -6$ The **product** of two numbers is the answer when you **multiply** them. The product of 2 and -9 is -18 .**Careful.** “Two negatives make a positive” is only true for $\times$ **and** $\div$. For **adding** it is false: $-3 + -4 = -7$, because adding a negative still sends you **left**.

A1 Get the sign first

(a) Complete the four rules. Write **positive** or **negative** in each gap.(i) positive $\times$ positive is _____(iii) negative $\times$ positive is _____(ii) positive $\times$ negative is _____(iv) negative $\div$ negative is _____

(b) **Do not work out the digits.** Write only + or – for the sign of each answer.

(a) -13×-8 _____ (c) -25×16 _____ (e) 99×-99 _____

(b) $47 \div -1$ _____ (d) $-144 \div -12$ _____ (f) $-360 \div 9$ _____

A2 Work them out

Now do the digits as well as the sign.

(a) $-7 \times 3 =$ _____ (e) $-10 \times 8 =$ _____ (i) $56 \div -7 =$ _____

(b) $-8 \times -4 =$ _____ (f) $0 \times -23 =$ _____ (j) $-72 \div -8 =$ _____

(c) $9 \times -6 =$ _____ (g) $-45 \div 5 =$ _____ (k) $-100 \div 4 =$ _____

(d) $-5 \times -11 =$ _____ (h) $-36 \div -6 =$ _____ (l) $27 \div -27 =$ _____

A3 Say it, then write it

Word help – these words decide the order of your calculation

multiply –2 by 5 -2×5 .

the product of the answer you get when you **multiply**. The product of -3 and -2 is 6.

divide –18 by 3 $-18 \div 3$. The number being **shared out** is -18 .

3 divided into –18 the **same** calculation, $-18 \div 3$. The English swaps the two numbers round; the calculation does not.

share between divide it out equally.

a debt money you must pay back. It makes a total **negative**.

each / every / per one lot, repeated – and repeated adding is **multiplying**. So **falls by 4 every hour** means each hour the change is -4 .

Turn each English sentence into a calculation, then work out the answer. **Write the calculation first.**

1. Multiply -6 by 7.

Calculation: _____ Answer: _____

2. Find the product of -4 and -9 .

Calculation: _____ Answer: _____

3. Divide -56 by 8.

Calculation: _____ Answer: _____

4. Work out 5 divided into -40 .

Calculation: _____ Answer: _____

5. The temperature falls 4 degrees every hour for 7 hours. What is the total change?

Calculation: _____ Answer: _____

6. A debt of 60 dollars is shared equally between 4 people. Write the amount **each** person owes as a negative number.

Calculation: _____ Answer: _____

A4 Brackets first

Remember: the new rule does not live inside the bracket

Work out **what is inside the bracket** first. Inside it you are often **adding**, and adding a negative still sends you **left**: $(-3 + -2) = -5$, **not** 5.

(a) Work these out. Show what is inside the bracket first.

(i) $(-3 + -5) \times 4 =$ _____

(iv) $(-9 - -3) \div 2 =$ _____

(ii) $(2 - 7) \times -6 =$ _____

(v) $-8 \times (-2 + 2) =$ _____

(iii) $(-4 + -6) \div -5 =$ _____

(vi) $(10 + -4) \times -3 =$ _____

(b) A student wrote $(-3 + -5) \times 4 = 32$.

In one full sentence, explain which rule they have used in the wrong place.

A5 Catch it with an estimate

Remember: an estimate is a rough answer you work out first

Estimate: round each number to something easy, then calculate. If your real answer is nowhere near it, you have made a mistake. Example: -4.1×2.8 rounds to $-4 \times 3 = -12$.

(a) Mr Bowen works out -4.9×21 and gets 102.9. Estimate the answer by rounding both numbers. Estimate: _____

(b) His answer must be wrong. Say what he has forgotten.

A6 Word problems

Read each one **all the way to the end** before you start writing. Show the calculation, then the answer.

1. The quiz game. In Mr Bowen's quiz game, every wrong answer scores -15 points.

- (a) A student gets 6 answers wrong. What is the change to their score?
- (b) In a later game, a student's score changes by -120 points from wrong answers alone. How many did they get wrong?

2. Eight forgotten debts. Mr Bowen owes 8 different friends 7 dollars each.

- (a) Write his total debt as a negative number.
- (b) All 8 friends then tell him to forget the debt, so all 8 debts of -7 dollars are **taken away** from his total. Write this as a multiplication and work out the change.
- (c) Is Mr Bowen better off or worse off than before? By how much?

3. Mr Bowen sits his own quiz. A correct answer scores 4 points and a wrong answer scores -4 points. Mr Bowen answers all 20 of his own questions. He gets 10 right and 10 wrong.

- (a) How many points does he get from the correct answers?
- (b) How many points does he get from the wrong answers?
- (c) What is his total score?

A7 Now you write the question**Your turn to be the teacher**

So far the questions have given you the English, and you have written the maths. Now do it **backwards**. You are given the maths – write the English.

Your word problem must:

- be about something **real** – money, temperature or points in a game;
- use a word from the **Word help** box, such as **each**, **every**, **per**, **shared between** or **a debt**;
- be written in **full English sentences**, and end with a question;
- have the answer that is already given to you.

(a) Write a word problem for

$$5 \times -8 = -40$$

(b) Write a word problem for

$$-60 \div 12 = -5$$

Section B • Science 1.2 – Animal Cells**Remember from the lesson**

You are an animal. An animal cell has a **cell membrane**, **cytoplasm**, **mitochondria** and a **nucleus**. It has **no cell wall**, so it has **no fixed shape**. A plant cell has all of those **plus** a **cell wall**, **chloroplasts** and a **sap vacuole**.

B1 The words for comparing two things

Word help – how to compare in English

similar	alike in some ways, different in others.
the same	no differences at all. It is stronger than similar .
Both A and B ...	the thing that follows is true of each of them.
..., but ...	joins two halves that disagree .
Unlike A, B ...	B is different from A in the way that follows.

Fill each gap with one of: **Both**, **and**, **but**, **Unlike**, **have**, **similar**, **the same**. Each word is used once.

- _____ a plant cell _____ an animal cell have a nucleus.
- A plant cell has chloroplasts, _____ an animal cell does not.
- _____ a plant cell, an animal cell has no fixed shape.
- Both kinds of cell _____ mitochondria.
- An animal cell is _____ to a plant cell, but the two are not _____.

B2 True or false?

Tick one box for each sentence. Then **rewrite each false sentence** on the lines below so that it becomes true.

Sentence	True	False
1. Every animal cell has a nucleus.		
2. An animal cell has a cell wall made of cellulose.		
3. Mitochondria are found in plant cells and in animal cells.		
4. An animal cell has no fixed shape because it has no cell wall.		
5. A plant cell has three parts an animal cell has not.		
6. Chloroplasts are green because they contain cellulose.		

Rewrite the false sentences here, in full:

B3 Answer in full sentences**Sentence starters – you may copy these**

“Every animal cell has...”

“An animal cell has no fixed shape because...”

“Both a plant cell and an animal cell have..., but only...”

“Unlike a plant cell, an animal cell...”

1. Name the **four** parts that every animal cell has.

2. A plant stands still, but an animal moves – it bends, squeezes and changes shape all day. Explain why an animal cell has **no fixed shape**. Use the word **because**.

3. Give **two** ways an animal cell is different from a plant cell. Write each one as a full sentence, using **but** or **Unlike**.

B4 Two pictures down a microscope

Mr Bowen looks at two slides. **Picture 1** shows cells packed tightly together like bricks in a wall, each one with straight, stiff edges. **Picture 2** shows soft, rounded cells with no straight edges anywhere.

(a) For each picture, say whether it shows **plant** cells or **animal** cells, and say **how you know**. Use the word **because**.

Picture 1:

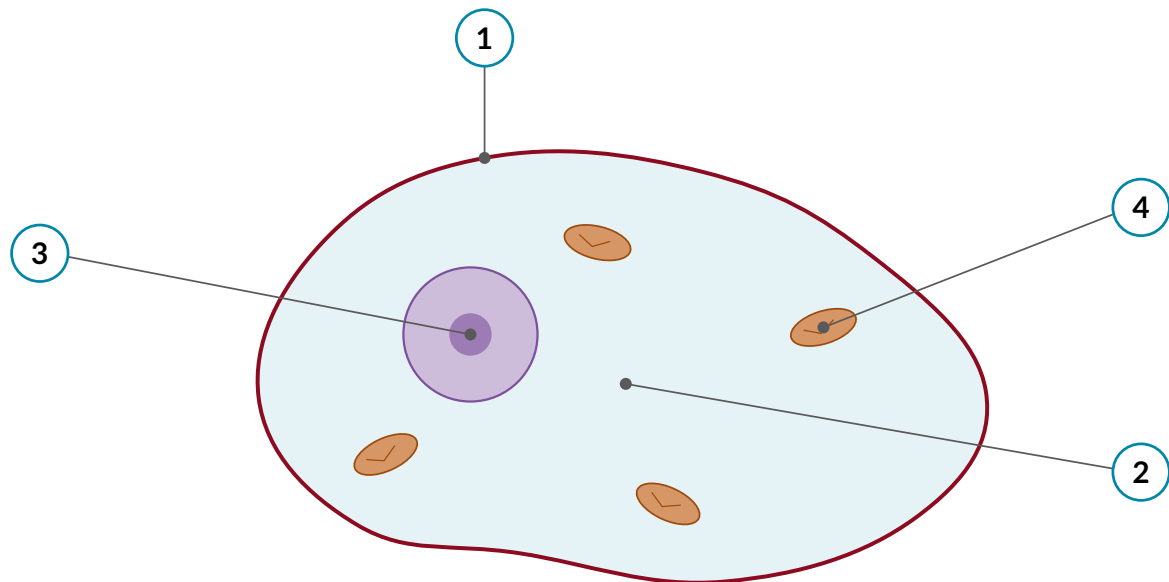
Picture 2:

(b) Neither picture has anything green in it. A student says: “There is no green, so both pictures must show animal cells.”

Explain in one full sentence why the student is wrong. (Hint: think about where an onion grows.)

B5 Label the animal cell

Write the name of each numbered part on the lines below the diagram.



1. _____

3. _____

2. _____

4. _____

(b) Name the **three** parts you would have to **add** to this diagram to turn it into a **plant** cell.
